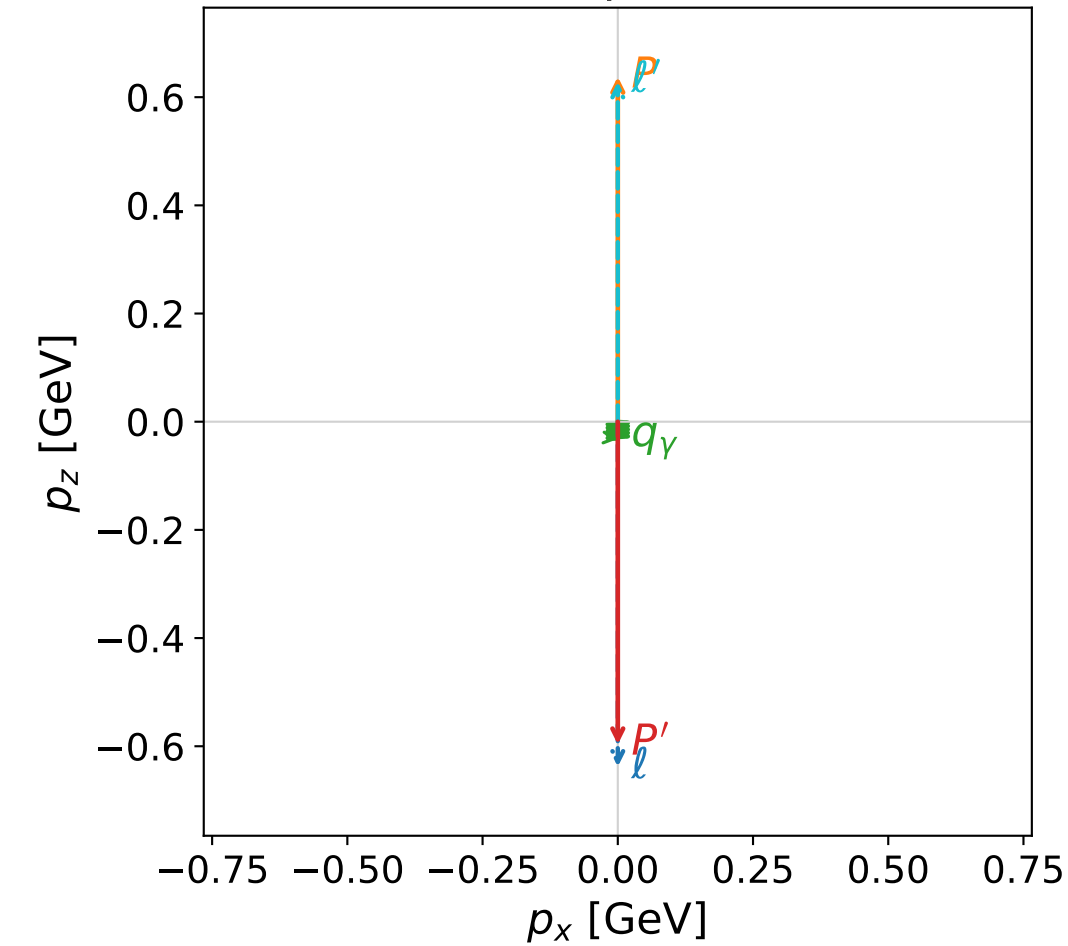


# coherent mixing angles: maximum $C_{ep}$ configuration (gradient\_local\_minimum)

z--x plane



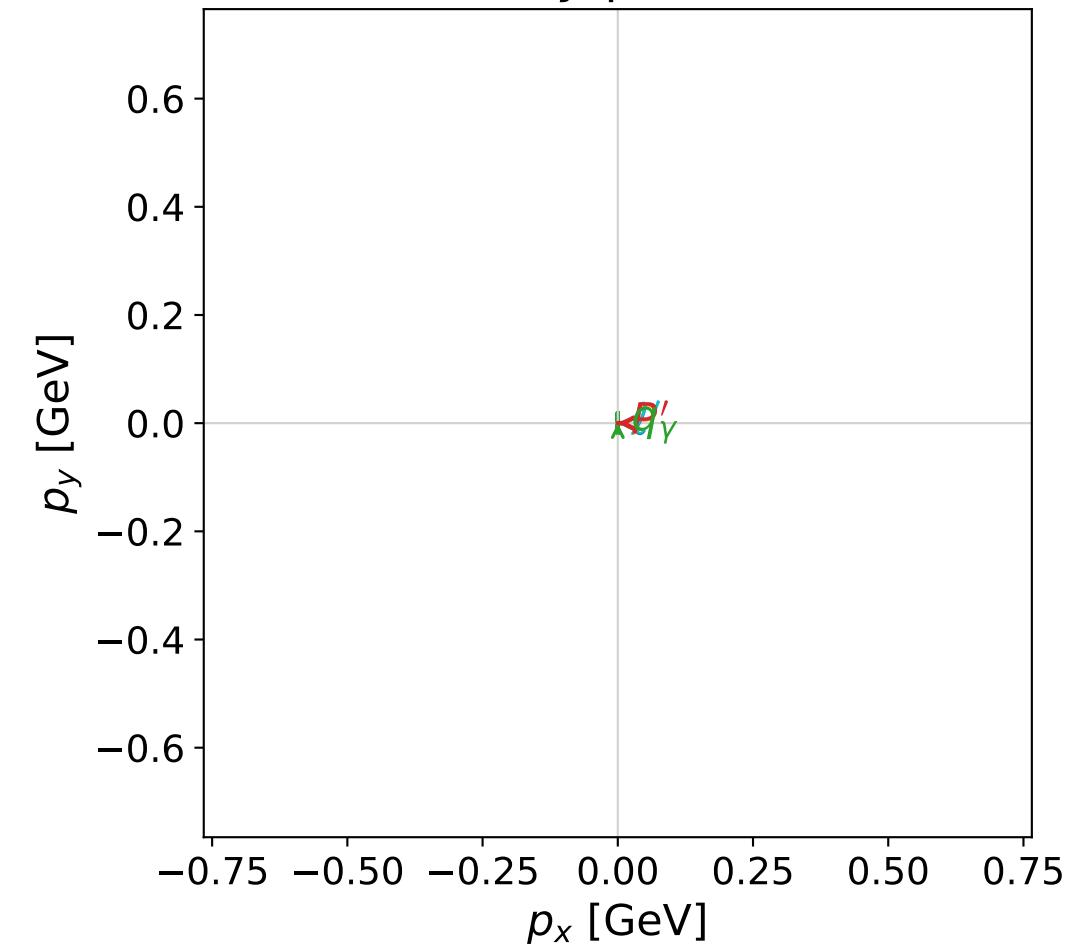
max\_c\_ep\_mixing\_angles\_polarization\_02\_local\_minimum\_16  
 $C_{ep}=1$   
 region: polarization\_cluster\_02\_local\_minimum\_16  
 outgoing-state purity: 1  
 kinematic point: gradient\_local\_minimum  
 lepton mixing angle:  $\alpha_e=0.44786895$  rad  
 incoming lepton:  $0.901372|+\rangle + 0.433046|-\rangle$   
 proton mixing angle:  $\alpha_p=2.0527958$  rad  
 incoming proton:  $-0.463552|+\rangle + 0.88607|-\rangle$

$s=3.14088$ ,  $\sqrt{s}=1.77225$ ,  $|\vec{P}|=0.637899$ ,  $|\vec{P}'|=0.596715$   
 $\theta_{p'}=3.14159$ ,  $\theta_\gamma=3.14099$ ,  $\phi_{p'}=3.12666$ ,  $\phi_\gamma=6.28293$   
 $E_\gamma=0.031911$ ,  $\theta_{l'}=3.05541e-05$ ,  $\phi_{l'}=3.14133$   
 $Q^2=1.604$ ,  $x_B=0.97992$ ,  $t=-1.52376$ ,  $W^2=0.912712$ ,  $y=0.723945$

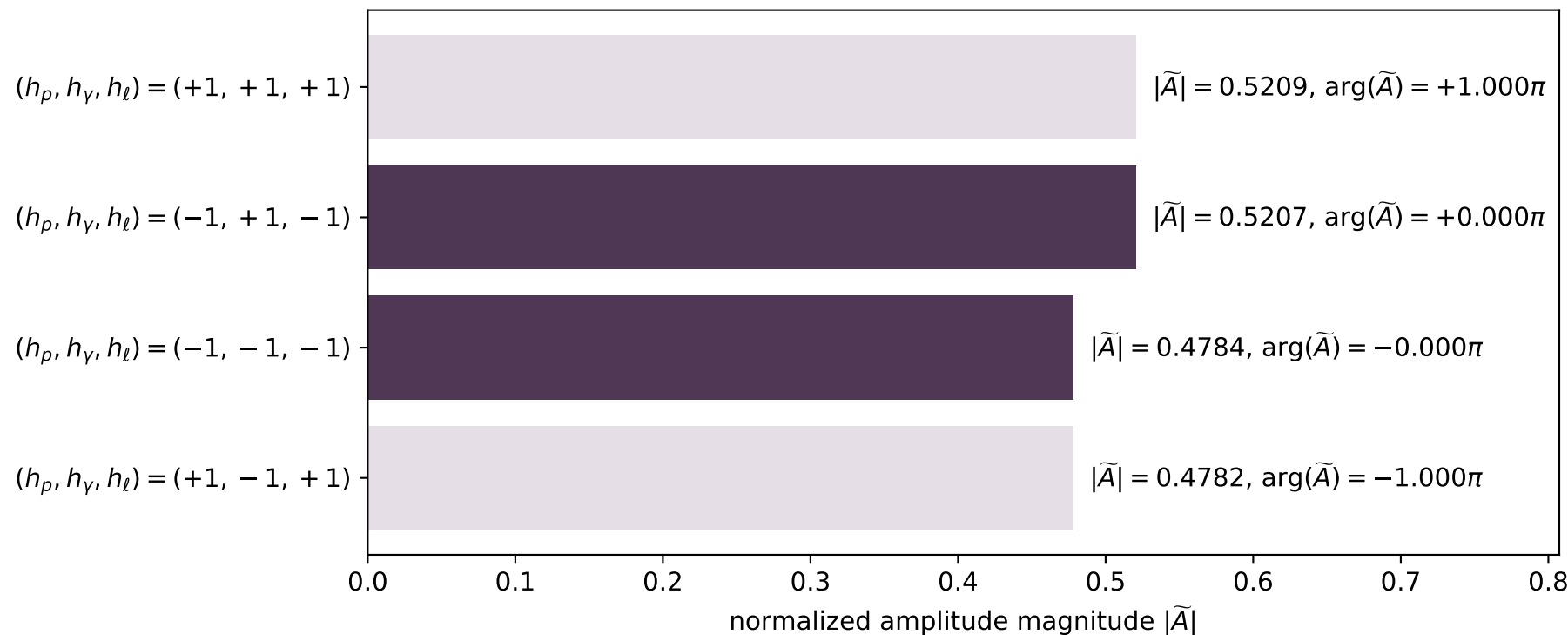
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|            |      |        |        |         |        |         |        |         |
|------------|------|--------|--------|---------|--------|---------|--------|---------|
| $l$        | $E=$ | 0.6379 | $p_x=$ | 0.0000  | $p_y=$ | 0.0000  | $p_z=$ | -0.6379 |
| $P$        | $E=$ | 1.1344 | $p_x=$ | 0.0000  | $p_y=$ | 0.0000  | $p_z=$ | 0.6379  |
| $l'$       | $E=$ | 0.6286 | $p_x=$ | -0.0000 | $p_y=$ | 0.0000  | $p_z=$ | 0.6286  |
| $P'$       | $E=$ | 1.1117 | $p_x=$ | -0.0000 | $p_y=$ | 0.0000  | $p_z=$ | -0.5967 |
| $q_\gamma$ | $E=$ | 0.0319 | $p_x=$ | 0.0000  | $p_y=$ | -0.0000 | $p_z=$ | -0.0319 |

x--y plane



Leading normalized final-state amplitudes  
(N=4, retained=100.0%)



electron: polarization cluster P2, local minimum 16; pairwise projections of the 8D  $1 - C_{ep} = (1 - C_{ep})_{\text{local}} + 0.05$  contour

